Chittagong University of Engineering & Technology
Department of Computer Science & Engineering
B. Sc. Engineering Level-III, Term-II, Exam. 2023

Course No: CSE-311
Course Title: Computer Networks
Marks: 210
Time: 3 Hours

The figure in the right margin indicates full marks. The questions are of equal value. Answer any three questions from each section. Use separate script for each section.

SECTION-A

Q.1(a) The OSI and TCP/IP models both define network communication but differ in structure and approach. If you had to design a new protocol stack, what key elements would you take from each model and why? Justify your answer with practical examples. 10

Q.1(b) Describe the significance of "Access Network". With the rise of streaming and remote work, access network technology is best suited for future demands. Consider aspects such as bandwidth, latency and scalability in your answer. 10

Q.1(c) What do you mean by protocol, topology, and collision domain? Explain how network collision can be handled with CSMA/CD and CSMA/CA. 13

Q.2(a) What do you know about wired and wireless networking? Discuss different types of network topologies on computer network. 10

Q.2(b) What do you mean by framing and flow control? Compare among different framing and flow control methods. 12

Q.2(c) What is hamming distance? Find the hamming codes for the alphabets of the word "**CAFE**" consider the ASCII codes: **C → 1000011, A → 1000001, F → 1000110, E → 1000101**. 13

Q.3(a) Write down the algorithm for computing Checksum. Calculate the Check summed frame that is to be transmitted for the input frame **101101110011**. Use the generator $G(X) = X^5 + X^2 + 1$. 10

Q.3(b) Explain the concept of bit stuffing. You receive the following data fragment: 10

0110 0111 1100 1111 0111 1101

You know that the protocol uses bit stuffing. Show the data after destuffing.

Q.3(c) Distinguish between 15

(i) Port and socket number
(ii) Packet and frame
(iii) Cloud computing and quantum computing
(iv) TCP and UDP (V) VPN and VLAN

Q.4(a) Design a client-server model and peer-to-peer communication-based network model for the following: 10

(i) A mobile application with IoT and cloud computing
(ii) Autonomous vehicles

Q.4(b) What do you mean by IP addressing and subnetting? With proper example distinguish between 12

(i) Public and Private IP addresses
(ii) IPv4 and Ipv6

Q.4(c) Find subnet of **172.16.2.1/22** which will be applicable for your office room having 50 Pcs and 23 Pcs. Also find the first and last usable IP address along with broadcast address. 13

SECTION-B

Q.5(a) A financial trading system requires low latency and high-reliability network communication. Would you recommend a datagram or virtual-circuit network model for this application? Justify your answer with a detailed analysis of latency, reliability and fault tolerance. 12

Q.5(b) Consider the network of Figure 5 (b). Distance vector routing is used, and the following vectors have just come into router C: 13

From B: (5, 0, 8, 12, 6, 2), from D: (16, 12, 6, 0, 9, 10), and from E: (7, 6, 3, 9, 0, 4). The cost of the links from C to B, D and E are 6, 3, and 5, respectively. Calculate the C's new routing table? Give both the outgoing line and the cost.

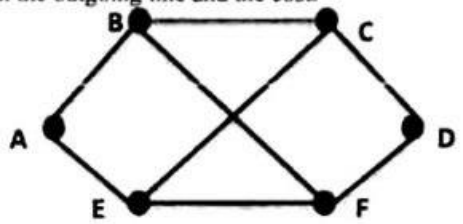

Figure 5 (b)

Q.5(c) Explain the significance of Maximum Transmission Unit (MTU). Explain the process of path MTU discovery. 10

Q.6(a) Describe the main drawbacks of distance vector routing. How do link state routing overcomes this drawback? 10

Q.6(b) Define default route, sink tree, route poisoning, and split horizon with proper example. Use Dijkstra's algorithm for finding the shortest path from A to F for the following network (Figure 6 (b)). 12

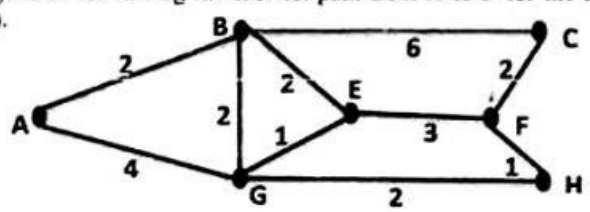

Figure 6 (b)

Q.6(c) Write down the prominent factors of congestion in a network. Briefly explain leaky bucket, token bucket and hop by hop choke packet algorithm for congestion control. 13

Q.7(a) What is ACL? How explicitly deny works in ACL? What do you mean by extended and standard ACL? What are the features of 5G (communication) technology? 12

Q.7(b) Why do we need cryptography? Briefly explain different cyber-attacks, substitution ciphers and transposition ciphers. 10

Q.7(c) Write down the principles of 'Date Encryption Standard (DES)' algorithm for network security. By using RSA algorithm encrypt the words '**COMPUTER**'. 13

Q.8(a) Explain the role of SMTP, PoP3, and IMAP in email communication. What are the advantages and disadvantages of each method when it comes to retrieving and managing emails. 10

Q.8(b) Explain the process of assigning IP addresses using DHCP. What is the significance of DHCP leasing and DHCP renewal? 10

Q.8(c) Consider the IPv4 packet shown in Figure 8 (c), that a router received. The Maximum Transmission Unit (MTU) of the outgoing link is 500 bytes. 15

(i) Can the router forward the packet? Justify your answer.

(ii) Does the packet require fragmentation? Can the router fragment the packet? Why or why not?

(iii) If the packet is dropped what ICMP message should the router send back to the sender? Include the type and reason for the message.

(iv) If fragmentation were allowed, how many fragments would be created and what would their offset values be?

IPv4	20 (byte)	63		1400 (bytes)		
254				1	0	0
5						
192.168.1.100						
10.10.2.25						

Department of Computer Science and Engineering
B. Sc. Engineering L-3, Term-II, Exam. 2022

Course No: CSE-311
Course Title: Computer Networks
Marks: 210
Time: 3 Hours

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The figure in the right margin indicates full marks. The questions are of equal value. Answer any three questions from each section. Use separate script for each section.

Section-A

Q.1(a) What is the formal definition of Internet? Design a client-server model and peer-to-peer communication model for the followings: 10
- i) A mobile application using cloud computing.
- ii) Autonomous vehicles.

Q.1(b) What do you know about protocols and topologies? Discuss different types of network topologies on computer network. 12

Q.1(c) Write short notes on: 13
- i) Collision domain and broadcast domain.
- ii) Hub, switch, router, and gateway.
- iii) Token passing and FDDI
- iv) Ethernet, ATM, cloud computing.

Q.2(a) A system has a 5-layer protocol hierarchy. Application generate messages of length m bytes. At each of the layers, respective header is added. Show the different phases of the message during the delivery from one end system to another through a link layer switch and a router. 10

Q.2(b) Consider the signal shown in Fig. 2(b). Such signal is being transmitted through a noiseless channel of capacity 128 Kbps (maximum). 13
- i) Calculate the bit rate of the signal.
- ii) What is the bandwidth of the signal?

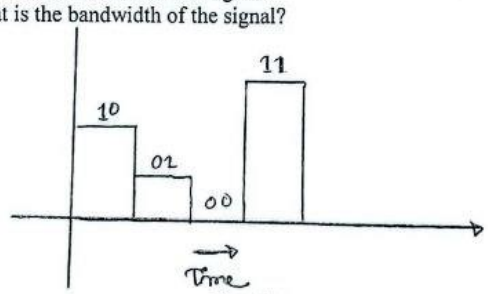

Fig. 2(b)

Q.2(c) Describe the process of computing checksum. Calculate the checksummed frame that is to be transmitted for the input frame 101101110011. Consider the generator $G(x) = x^5 + x + 1$. Use CRC algorithm. 12

Q.3(a) How does TCP connection establish and terminate? Compare among TCP and UDP protocols. 10

Q.3(b) What do you mean by 1-persistant CSMA, non-persistent CSMA, and hidden station problem? How collision can be handled with CSMA/CD and avoided with CSMA/CA mechanism? 12

Q.3(c) What do you mean by PPP and HDLC? Compare among: 13
- i) Port and socket number.
- ii) Segment, packet and frame.
- iii) Bit stuffing, byte stuffing, and character count method of framing.
- iv) Go-back N, selective repeat, stop and wait (sliding window) ARQ protocol.

Q.4(a) Describe the TCP 3-way handshake and sliding window protocol for flow control. 12

Q.4(b) Give a taxonomy showing various options for connecting subscribers to the WAN. Explain the steps perform by router upon arrival of a packet. Also discuss the way of building MAC table by a switch. 08

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Q.4(c) Why do we need VLSM and subnetting? Give a range for private IP address. 07

Q.4(d) Find subnet of 172.16.2.1/22 which will be applicable for your office room having 50 PCs and 23 PCs. Also find first and last usable IP address along with broadcast address? 08

Section-B

Q.5(a) Explain the difference between routing, forwarding, and switching. 06

Q.5(b) A router has the following (CIDR) entries in its routing table: 12

Address/mask	Next hop
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Router 1
default	Router 2

For each of the following IP addresses, what does the router do if a packet with that address arrives?

i) 135.46.63.10
ii) 135.46.57.14
iii) 135.46.52.2
iv) 192.53.40.7
v) 192.53.56.7
vi) 192.53.56.255

Q.5(c) Explain the main drawback of Distance Vector routing. 08

Q.5(d) Describe the concept of Tunneling with appropriate figures. 09

Q.6(a) What are the main features of IPV6? Draw the header formats of IPV6 and describe the fields. 10

Q.6(b) What are the principles of security in computer network? Discuss different active and passive security attacks. What do you mean by Caesar cipher and Vigenere cipher. 10

Q.6(c) If the source and destination PCs are in two separate networks, how the source PC can obtain the MAC address of the destination PC? Explain with necessary diagram. 10

Q.6(d) Write a short notes on 6G technology, and blockchain. 05

Q.7(a) Point out the main factors of congestion in a network. In handling congestion, what are the main drawbacks of "Warning bit" mechanism? Describe the approach you think suitable to overcome the drawback of warning bit mechanism. 12

Q.7(b) Briefly describe the five sequential steps of link state routing. 08

Q.7(c) What do you mean by internet control protocol? What are the main internet control protocols? 05

Q.7(d) Suppose you connect your PC to the wireless network at your home. The wireless network runs DHCP to allocate IP. Explain the procedure of allocating and maintaining IP address for your PC in this network. 10

Q.8(a) With the help of appropriate flowchart, explain the operation principles of inbound and outbound ACLs. 10

Q.8(b) With necessary diagram briefly explain the symmetric key algorithm "Data Encryption Standard (DES)". 15

Q.8(c) Write short notes on: 10

i) POP3
ii) SMTP
iii) HTTP
iv) FTP

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Chittagong University of Engineering and Technology
Department of Computer Science and Engineering
B. Sc. Engineering Level-3, Term-II, Exam. 2021

Course No: CSE-311
Course Title: Computer Networks
Marks: 210
Time: 3 Hours

The figure in the right margin indicates full marks. The questions are of equal value. Answer any three questions from each section. Use separate script for each section.

SECTION-A

Q.1(a) What is computer network? Distinguish between 09
- i) Computer vs. autonomous systems
- ii) Client server model vs. Peer-to-peer communication model

Q.1(b) What is Ethernet? Discuss different types of network topologies in computer network. 10

Q.1(c) What do you know about 08
- i) Wired and wireless networking
- ii) Collision domain and broadcast domain

Q.1(d) Describe the construction and operation of 08
- i) Hub, switch and router
- ii) Token passing and FDDI

Q.2(a) What do you mean by framing? How collision can be handled with CSMA/CD and avoided with CSMA/CA mechanism? 09

Q.2(b) Which OSI layer is used for the following 08
- i) To route packets
- ii) To run services like FTP and Telnet
- iii) To detect and correct error
- iv) To covert packets to frame.

Q.2(c) Discuss why sliding window protocol is used for flow control. Compare among 09
- i) Bit stuffing and byte stuffing methods of framing
- ii) Go back-N and selective repeat ARQ protocol

Q.2(d) Find the hamming code for the word "SEE", consider the following ASCII codes: 09

S: 01010011, E: 01000101

Q.3(a) Describe the TCP 3-way handshaking and TCP congestion control scheme in brief. 13

Q.3(b) Describe the checksum algorithm for error detection. Calculate the checksummed frame that is to be transmitted for the input frame 101101110011. Consider the generation $G(X) = X^5 + X + 1$. Use CRC algorithm for error detection 10

Q.3(c) Distinguish between 12
- i) Port and socket number
- ii) VLAN and WAN
- iii) IoT and cloud computing

Q.4(a) What do you mean by IP addressing, subnetting and VLSM? With proper example, distinguish between public and private IP address. 11

Q.4(b) Consider you are given the address space for CUET 172.16.10.0/23. Now, your task is to subnet the address according to the hos requirements for different departments that are given in the following table. Fill up the empty columns with values resulting from your calculation. Use the smallest subnet sizes that will accommodate the relevant number of hosts. 15

Department	No. of hosts required	Network address	Broadcast address	Subnet mask
CSE	125			
EEE	150			
ETE	60			
MIE	20			
BME	50			

Q.4(c) Why IP is called the connectionless protocol? Compare among transparent and non-transparent segmentation. 09

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SECTION-B

Q.5(a) What are the two major classes of routing algorithm? What do you mean by routing and forwarding? Why do we need distance vector and link state routing protocol? 09

Q.5(b) What is congestion in computer network? Briefly explain leaky bucket and token bucket algorithm for congestion control. 09

Q.5(c) How congestion can be controlled using hop-by-hop choke packet? 08

Q.5(d) Define count-to-infinity problem of distance vector routing algorithm. What do you mean by static and dynamic routing? 09

Q.6(a) Define default route and sink tree with proper examples. Use Dijkstra's algorithm for finding the shortest path from A to F for the following network. 10

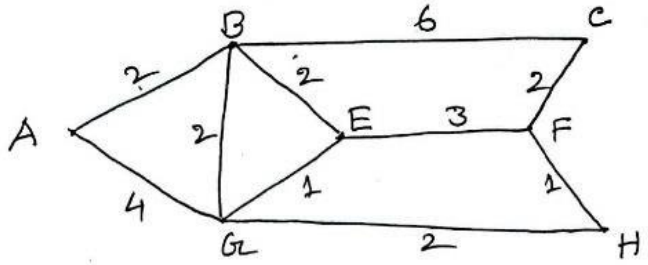

Fig. 6(a)

Q.6(b) What is 08
- i) Route poisoning and split horizon?
- ii) Loadshedding and QoS

Q.6(c) What is ACL? How explicit deny works in ACL? What do you mean by extended and standard ACL? 09

Q.6(d) Draw the IP datagram header format by indicating the size of each field. 08

Q.7(a) What do you understand by DNS namespace? How DNS resolver looks up a remote name in 8 steps? 10

Q.7(b) Suppose cse.cuet.ac.bd and ete.cuet.ac.bd are under the same local DNS server. Considering no initial DNS caching along the path, answer the following questions with appropriate figures 15
- i) Show the DNS resolver steps when resolver on cse.cuet.ac.bd wants to know the IP address of cs.hbs.edu
- ii) After the access of cse.cuet.ac.bd, ete.cuet.ac.bd also wants to access cs.hbs.edu. Show the steps of this later access.

Q.7(c) Write short notes on 10
- i) SSH
- ii) FTP
- iii) SMTP
- iv) WWW
- v) HTTPs

Q.8(a) Why do we need cryptography? What is cipher? Briefly, explain substitution and transposition ciphers? 12

Q.8(b) Write down the overall operation of RSA algorithm. By using RSA, encrypt the word "COMPUTER". 13

Q.8(c) A key of transposition cipher is "DICTIONARY" and the plaintext is "happiness is temporary". Find the encrypted text. 10

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Chittagong University of Engineering and Technology
Department of Computer Science and Engineering
B. Sc. Engineering Level-3, Term-II, Exam. 2020

Course No.: CSE-311
Course Title: Computer Networks
Marks: 210
Time: 3 Hours

The figure in the right margin indicates full marks. The questions are of equal value. Answer any three questions from each section. Use separate script for each section.

SECTION-A

Q.1(a) Why Internet is called a network of networks? Justify. What is the relation between Internet and compute network? What are the components of computer network? 8

Q.1(b) What do you know about unicast, broadcast, and multicast mode of communication? Compare among different types of networks topologies in computer network with their pros and cons. 10

Q.1(c) Distinguish between collision domain and broadcast domain. How is a switch different from a router? 7

Q.1(d) Write short notes on the following: 10
(i) IoT, cloud computing, and distributed systems,
(ii) Gigabit Ethernet, ATM, Token ring, and FDDI,
(iii) Wired medium and wireless medium.

Q.2(a) Describe the layers of TCP/IP reference model with their respective functionalities. Why is TCP/IP more popular than OSI reference model? 9

Q.2(b) What do you mean by framing? Describe character count and bits stuffing methods of framing with their advantages and disadvantages. 8

Q.2(c) Describe the TCP 3-way handshaking and TCP congestion control scheme in brief. 10

Q.2(d) Describe the checksum algorithm for error detection. Use that algorithm for calculating the check-summed frame that is to be transmitted for the frame **101100101**. Consider the generator $x^3 + x^2 + 1$. Use CRC algorithm for error detection. 8

Q.3(a) Compare among segment, packet, and frame. Discuss the features of TCP and UDP. 7

Q.3(b) What do you mean by port and socket number? Briefly explain why sliding window protocol is used for flow control. 8

Q.3(c) How Hamming code can be used for error correction? Find the Hamming code for the word "CSE". Consider the following ASCII codes: 10
C: 01000011, S: 01010011, E: 01000101.

Q.3(d) Distinguish among 10
(i) CSMA/CD and CSMA/CA
(ii) Twisted pair and fiber optic cable,
(iii) Datalink layer and network layer,
(iv) 1-persistent CSMA and non persistent CSMA.

Q.4(a) What is IP addressing? Distinguish between 10
(i) Classful and Classless IP addressing,
(ii) Subnetting and. Supernetting,
(iii) Subnet mask and wildcard mask.

Q.4(b) Consider 'CUET IT Business Incubator' has a block of IP addresses **210.210.210.0/24**. We need to create as many subnets so that each subnet contains minimum of 15 usable IP addresses. 10

(i) How many host bits should be borrowed as network bits?
(ii) How many usable subnets will be created?
(iii) What will be the network address of last usable subnet?
(iv) What will be the second usable IP address of the second usable subnet?

Q.4(c) Why IP is called connectionless protocol? Discuss why do we need subnetting and VLSM? 7

Q.4(d) Compare among IPV_4 and IPV_6 features. Why do we need private IP address and loopback test? 8

SECTION-B

Q.5(a) What is routing protocol in computer network? Briefly explain the features of (i) distance vector and link state routing protocol, (ii) Intra domain and inter domain routing protocol. 10

Q.5(b) Define default route and sink tree with appropriate examples. Use Dijkstra's algorithm for finding the shortest path from A to D for the following network. 14

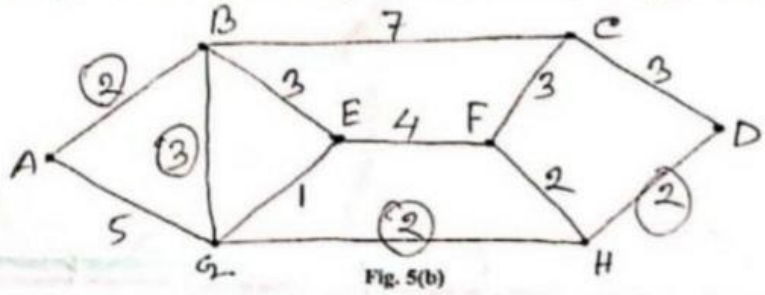

Fig. 5(b)

Show the steps for calculation. How can you modify the Dijkstra's algorithm to achieve better result?

Q.5(c) What is route poisoning and split horizon? 6

Q.5(d) Compare among ACL and Firewall. 5

Q.6(a) What are the major factors of congestion in a network? Discuss the approaches of congestion control in virtual circuit subnet and in datagram subnet. 15

Q.6(b) Why Hop by hop choke packets are better than a choke packet that affects only the source? Describe with appropriate figure. 10

Q.6(c) What is fragmentation? Describe transparent and non-transparent fragmentation along with their advantages. 10

Q.7(a) What do you understand by DNS Name space? Describe how a resolver looks-up the remote name? Consider a resolver on **eecs.admission.mit.edu** wants to know the IP address of the host **cuet.ac.bd**. 10

Q.7(b) What are the major drawbacks of SMTP protocol? Briefly describe the operations of POP-3 protocol. 10

Q.7(c) Write short notes on: 10
(i) ARP,
(ii) DHCP.

Q.7(d) Why Hub is called a device with no intelligence? 5

Q.8(a) Differentiate between 10
(i) Private key and Public key encryption,
(ii) Symmetric key and Public key signature.

Q.8(b) A key of the transposition chipper is **"LOCKDOWN"** and the plain text is **"stayhomestaysafe"**. Find the encrypted cipher text. 10

Q.8(c) Describe the overall operation of DES. 15

-The End-

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Chittagong University of Engineering and Technology
Department of Computer Science and Engineering
B. Sc. Engineering Level-3, Term-II, Exam. 2019

Course No.: CSE-311
Course Title: Computer Networks
Marks: 210
Time: 3 Hours

The figure in the right margin indicates full marks. The questions are of equal value. Answer any three questions from each section. Use separate script for each section.

Section-A

Q.1(a) What are the types of **OSI** reference model? Give a comparative analysis between **OSI** reference model and **TCP/IP** model. 12

Q.1(b) What do you know about protocols and topologies? Write down the names of two logical topologies with their pros. and cons.? 11

Q.1(c) What is gigabit Ethernet? How collision handled with carrier sense multiple access with collision detection method (**CSMA/CD**)? 12

Q.2(a) Write down the reasons for using layered protocols. 8

Q.2(b) Write down the names of four methods that are used to mark the start and end of each frame. Give appropriate examples for any two methods. 9

Q.2(c) Given frame **10101011011** and generator **10001**. If receiver received **101010110101**, is it accurate reception? Proof by using cyclic redundancy check (**CRC**) method? 9

Q.2(d) State Shannon's theorem. A channel with a signal to thermal noise ratio (S/N) of 3 can transmit at a maximum rate of 4000 bps. Calculate the channel bandwidth. 9

Q.3(a) Differentiate between (i) **HTTP** and **HTTPS** (ii) **Telnet** and **SSL** (iii) **FTP** and **HTTP**. 09

Q.3(b) What is the importance of port number in communication? 05

Q.3(c) Differentiate between **UDP** and **TCP**. Also, explain three-way handshaking technique for connection establishment. 16

Q.3(d) How does **DHCP** work? 05

Q.4(a) What is IP addressing? How is it classified? How subnetting is performed? 08

Q.4(b) Where is the special IP address **127.0.0.0** used? What are the advantages of **CIDR** notation? 08

Q.4(c) What is $\mathbf{IPV_6}$. Explain its advantages over $\mathbf{IPV_4}$. Discuss the range of private IP address. 09

Q.4(d) Write short note on the following: (i) **Flow and error control** (ii) **Firewall** and **virtual LAN (VLAN)** (iii) **Intelligent Switch** and **router**. 10

Section-B

Q.5(a) Explain the concepts of broadcast domain and collision domain? Although, broadcast consumes huge bandwidth, there are some situations where we cannot ignore the necessity of broadcast. Explain one such 10

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situation.

Q.5(b) Suppose we have a block of IP addresses **100.100.100.0/24**. We want to create as many subnets as possible so that each subnet can provide at least **30 usable IP addresses.** Find the following: (i) Number of subnets that can be created, (ii) Last usable IP address of the last usable subnet, (iii) Subnet mask of the last usable subnet, (iv) Total number of usable IP address after subnetting. 10

Q.5(c) Differentiate between **Standard and Extended ACLs**. "It is necessary to place standard ACLs close to the destination while extended ACLs close to the source." Justify the statement with proper examples. 13

Q.6(a) Define two major classes of routing algorithms. Describe the **count-to-infinity** problem of distance vector routing algorithm. 10

Q.6(b) Explain some important features of hierarchical routing? What is default route? 12

Q.6(c) What is congestion in computer network? Briefly explain an algorithm for congestion control. 10

Q.7(a) With appropriate examples briefly explain substitution ciphers encryption and transposition ciphers. 13

Q.7(b) With necessary diagram, briefly explain the symmetric key algorithm "Data Encryption Standard (**DES**)". 12

Q.7(c) A key of the transposition cipher is "**QUESTION**" and the plain text is "**Weloveourcountrybangladesh**". Find the encrypted cipher text. 10

Q.8(a) What is fragmentation? Describe transparent and nontransparent *fragmentation along with their problems.* 15

Q.8(b) Write short notes on (i) **ICMP** (ii) **NAT** (iii) **SMTP** (iv) **Digital signature** (v) **FDDI**. 10

Q.8(c) Differentiate between (i) **Hub and Switch** (ii) **Bridge and router** (iii) **Internet and intranet** (iv) **Loopback address** and **physical address**.

-The End-

CSE-411

Chittagong University of Engineering and Technology
Department of Computer Science and Engineering
B. Sc. Engineering Level-4, Term-I, Exam. 2019

Course No.: CSE-411
Course Title: Computer Networks
Marks: 210
Time: 3 Hours

The figure in the right margin indicates full marks. The questions are of equal value. Answer any three questions from each section. Use separate script for each section.

SECTION-A

Q.1(a) What do you know about ATM virtual circuit? Briefly discuss some characteristics and benefits of ATM. 09

Q.1(b) With necessary diagram define client server model and peer-to-peer communication model. 06

Q.1(c) What is Ethernet? How does a token ring LAN operate? 12

Q.1(d) With the help of an appropriate figure show how information flows in a five-layer network. 08

Q.2(a) Circuit switching and packet switching differ in many ways. What are those differences? 10

Q.2(b) Write down the advantages and disadvantages of fiber optics over copper wire. 10

Q.2(c) State Nyquist's theorem. A noiseless channel can transmit binary signals at a maximum rate of 8600 bps. Find out the channel bandwidth. 10

Q.2(d) Draw the electromagnetic spectrum and show the working ranges of twisted pair cable and optical fiber. 05

Q.3(a) Describe the layers of TCP/IP reference model with their respective functionalities. Why TCP/IP model is more popular than OSI reference model? 15

Q.3(b) What do you mean by framing? Describe character count and bit stuffing methods of framing with their advantages and disadvantages. 10

Q.3(c) Compare among segment, packet, and frame. Discuss the features of TCP and UDP. 10

Q.4(a) Briefly explain why sliding window protocol is used for flow control. 10

Q.4(b) Describe the checksum algorithm for error detection. Use that algorithm for calculating the checksummed frame that is to be transmitted for the frame 101100101. Consider the generator $G(X) = X^3 + X^2 + 1$. 13

Q.4(c) What is Hamming distance? Find the Hamming code for the word "CUET". Consider the following ASCII codes 12

C: 0100 0011, U: 0101 0101, E: 0100 0101, T: 0101 0100

SECTION-B

Q.5(a) What is IP addressing? Distinguish between classful and classless IP addressing. 10

Q.5(b) Consider CUET has a block of IP addresses 210.210.210.0/24. We need to create as many subnets so that each subnet contains minimum of 15 usuable IP addresses 15

i. How many host bits should be borrowed as network bits?
ii. How many usuable subnets will be created?
iii. What will be network address of last usuable subnet?
iv. What will be second usuable IP address of second usuable subnet?

Q.5(c) Distinguish between subnetting and supernetting. What is the necessity of using loopback test? 10

Q.6(a) Write down the features of distance vector and link state routing protocol. 10

Q.6(b) Define sink tree with appropriate example. Use Dijkstra's algorithm for finding the shortest path from A to D for the following network (See figure-1). Show all the steps of calculation. 15

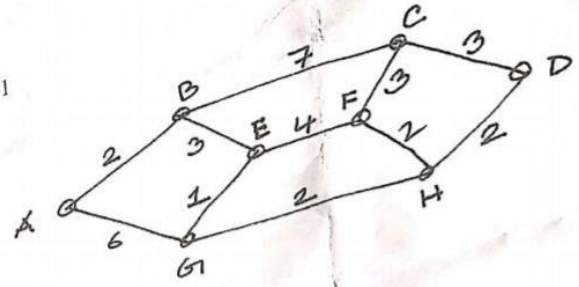

Figure: 6(b)

Q.6(c) Describe count-to-infinity problem of distance vector routing algorithm. 10

Q.7(a) What is congestion in computer network? Briefly discuss the *Hop-by-Hop choke packet* technique that is used for congestion control. 12

Q.7(b) What is VLAN? Write down the main features of IPv6. 12

Q.7(c) Define load shedding term used in congestion control. 05

Q.7(d) Write short notes on gateway and firewall. 06

Q.8(a) Distinguish between symmetric-key and public-key cryptosystem. Summarize the principle used in RSA cryptosystem. 13

Q.8(b) With appropriate examples briefly explain substitution ciphers encryption and transposition ciphers. 12

Q.8(c) A key of a transposition cipher encryption method is MEGABUCK and the plaintext is "pleasetransferfivelactakainmysonalibankaccount". Find the encrypted cipher text. 10

END

Chittagong University of Engineering and Technology
Department of Computer Science and Engineering
B. Sc. Engineering Level-3, Self Study, Exam. 2019

Course No: CSE-311/CSE-411
Course Title: Computer Networks
Marks: 210
Time: 3 Hours

The figure in the right margin indicates full marks. The questions are of equal value. Answer any three questions from each section. Use separate script for each section.

SECTION-A

Q.1(a) "We are living in a network centric world"- Justify the statement. 6

Q.1(b) Explain the roles of following layers of OSI reference model. 9
- i) Data link Layer
- ii) Session Layer
- iii) Transport Layer

Q.1(c) Explain the architecture of ATM reference model. 12

Q.1(d) Write down the concept of packet switching technology. 8

Q.2(a) What is topology of a network? Discuss different types of topologies used in LANs. What are the three criteria necessary for an effective and efficient network? 10

Q.2(b) Distinguish among segment, packet and frame. What are the advantages of using UDP over TCP? Which one is widely used and why? 9

Q.2(c) Explain how hamming code is used for error correction for a 4-bit data (1010). Discuss the procedure for TCP congestion control. 8

Q.2(d) Why do we need flow control and error control? In a wireless network, briefly explain the types of medium access control considering both hidden terminal and exposed terminal problem. 8

Q.3(a) Obtain the 4-bit CRC code work for the data bit sequence 10011011100 using the generator polynomial $x^4 + x^2 + 1$. 10

Q.3(b) Explain the checksum and two dimensional parity check procedure for error detection. 8

Q.3(c) How the inefficiency of stop-and-wait protocol is overcome in sliding window protocol? The bandwidth of a given channel is 1600 Hz. If the SNR is 20dB, what will be the maximum data rate of the channel in bps? 9

Q.3(d) What is gigabit Ethernet? Why do you require a limit on the minimum size of Ethernet frame? Compare between CSMA/CD protocol and CSMA/CA protocol. 8

Q.4(a) Differentiate between 18
- i) IPV_4 and IPV_6
- ii) Subnetting and VLSM
- iii) Public IP address and Private IP address
- iv) Network address and Broadcast address
- v) ARP and ICMP
- vi) NAT and PAT

Q.4(b) Write short notes on the followings: 12
- i) WWW and HTTPS
- ii) SMTP and FTP
- iii) Telnet and DHCP

Q.4(c) Why should we use port numbers in network communication? 5

SECTION-B

Q.5(a) Differentiate between
- i) Distance vector and Link state routing
- ii) Routing protocols and routed protocols
- iii) Static routes and dynamic routes
- iv) Stab networks and VLAN

12

Q.5(b) What is a routing loop? How can you avoid routing loops? Explain in any one method. 12

Q.5(c) What do you mean by hop count? Also, explain the concepts of autonomous number. 5

Q.5(d) Write short note on BGP. 6

Q.6(a) Explain how RSA algorithm can be used to encrypt the word "CUET". Given that prime numbers are P=11, Q=3. 8

Q.6(b) What are the four services required for secured communication. Explain the operation of the Diffie-Hellman protocol with example. 10

Q.6(c) Discuss the salient features and technologies of 5G communication system. 8

Q.6(d) What does minimum spanning tree protocol do to a switch? How does spanning tree determine priority? 9

Q.7(a) Explain the concept of collision domain and broadcast domain. "By using VLAN we can reduce the size of broadcast domain"- Justify. 13

Q.7(b) Differentiate between
- i) Single mode fiber and multimode fiber
- ii) Straight-through cable and cross-over cable
- iii) Coaxial cable and STP cable

12

Q.7(c) What is firewall? How will you apply security configuration in a router? 10

Q.8(a) Explain the procedure of building a switching table when a new switch is installed in a network. 15

Q.8(b) Consider that there are several access points installed in a campus network. In general, we need to login to each access point separately to be connected with the network. How can you ensure single login and get automatic access to all the access points? 8

Q.8(c) You are requested to plan a network backbone for a campus. What will be you prime considerations? List at least five of such considerations. 6

Q.8(d) Explain some situations where
- i) Wireless connections are more preferable than wired connections
- ii) Wired connections are more preferable than wireless connections

6